

ERA / Engineering Journal

Los 3 Mosqueteros · WRO Future Engineers 2026 Vocacional Manuel Méndez Liciaga · San Sebastián, Puerto Rico

Source: the supplied *Historial de documentación* notebook, through 2 October 2026. This is an English presentation of the team's existing account. Its observations and competition results are reported by the team; no new experimental results have been added. The original Spanish document and its spelling-corrected copy are indexed in [Docs](#).

Team

Name	Role recorded in the notebook
Adrián Iván Jiménez González	Programming
Carlos Elvin Cabán Martínez	Information and documentation
Joniel Emanuel Torres Traverzo	Vehicle design
Javier González Hernández	Teacher / coach

All student members represent Vocacional Manuel Méndez Liciaga in San Sebastián, Puerto Rico. The team name appears in the earlier notebook; the updated notebook names the robot ERA.

Design concept

The team initially sought a four-wheel car design that would be comfortable to work on. We took inspiration from a Formula 1-style layout: a compact structure suited to a smooth surface, with space for a simple two-motor arrangement and sensor placement. The goal was to organise the vehicle around a car-like drive structure.

The original notebook also describes a temporary three-wheel experimental configuration. That configuration created room to mount the sensors and reduced the possibility of cables interfering with wheels during early testing. It was a development prototype. The later supplied photographs show a four-wheel assembly.

March / research and initial construction

26 March 2026

The robot was handed over to the team. We received information about the competition rules, event locations and schedule.

27–28 March 2026

Research phase: we studied the LEGO SPIKE Prime programming environment and its programming methods.

30 March–2 April 2026

Major structural revision: we redesigned the robot into a simpler, more efficient and compact model to facilitate testing on the mat.

April / prototypes, sensors and navigation**6–7 April 2026**

We continued developing forward- and backward-motion logic. We tested it on a structurally simplified prototype dedicated to validating the program before further trials.

8 April 2026

Following the practice validation, we replaced and repositioned the motors. We converted the robot into a three-wheel prototype to make colour- and ultrasonic-sensor testing easier for obstacle-avoidance experiments.

9–10 April 2026

We updated the program for the new motor configuration and sensor positions, and reviewed obstacle detection and trajectory-following logic on the mat.

13–14 April 2026

We completed the sensor-code work for this stage, including trajectory recognition and route-following algorithms.

15 April 2026

Performance testing continued: speed calibration, trajectory recognition and sensor adjustments for obstacle avoidance.

16 April 2026

We organised the progress records, documenting the changes, iterations and updates made up to that date.

18 April 2026

The first national WRO 2026 practice competition took place in San Juan, Puerto Rico. The team could not attend because of personal circumstances.

13–22 April 2026 / work interruption

The updated notebook records an interruption caused by CRECE examinations and other academic commitments. This interval overlaps some separately recorded work entries; the source chronology has been retained rather than assigning new dates. The older notebook records a different limited-work interval, 19–27 April, and remains available in the archive.

23 April 2026

The colour sensor had difficulty reading the floor. We adjusted rotation when detecting the coloured floor lines and changed the robot's front structure to improve the operation of the downward-facing sensor. A software change was also made to improve stability in mat detection.

28 April 2026

We recalibrated the colour sensor for more precise recognition of the mat colours. The team reported easier corner navigation and lap completion after the adjustment.

29 April 2026

We updated the engineering notebook with the participant records and work schedule, reflecting the changes made to the robot.

May / national event and sensor limitations

1 May 2026

We finished the program changes for free running without obstacles. The team reported successful workshop practice in preparation for the national WRO competition.

2 May 2026 / event day one

The team attended the WRO competition. During the initial maintenance and programming period, we tested the program and its capabilities. Only minor changes were needed; no structural changes were made at that stage.

The notebook reports **28 points** in the first round. It describes two completed laps followed by a collision with a wall during the third lap. In workshop practice, the team had previously observed approximately five laps without that problem.

In the second round, the driving direction changed to counterclockwise. We worked on reversing the relevant values. The round also included a surprise rule and obstacles, requiring further trajectory and obstacle-avoidance work.

At that time, a scratched colour sensor affected recognition at a distance and made the new challenge difficult. Even if the program changes were functional, the defective sensor continued to limit performance. The notebook reports that the team ended day one with an 18-point advantage over the other teams. These are the team's recorded event observations, not newly recalculated scores.

3 May 2026 / event day two

The notebook records an opportunity to discard the worse of two rounds for the special challenge. The described task involved leaving a parking slot, completing a lap around the mat and returning to the parking area, with a random driving direction and obstacles.

We modified the program for that challenge without making structural changes. We worked on a program for both driving directions and obstacle avoidance, but the defective sensor continued to cause difficulties. This describes the local event as recorded in the notebook; it is not a replacement for the international 2026 challenge rules.

August / replacements and renewed testing

15 August 2026

Two team members attended a WRO competition briefing. We received replacement sensors to replace the defective one, as well as two spare ultrasonic sensors for future experiments.

26 August 2026

We tested variables including speed and rotation angles, and continued experimenting with the new sensors.

27 August 2026

We continued looking for alternatives to address the defective colour sensor within the limited spare-parts resources available in our workshop.

September / event feedback and structural improvements

5 September 2026

The team attended a second national WRO competition in San Juan, at the Coliseíto Pedrín Zorrilla. The notebook reports that the event rules remained the same as at the earlier competition.

We received feedback about a rule violation: obstacle detection, driving directions and challenge logic needed to be included in a single program slot. The notebook describes that slot as “0–1”; for an international SPIKE event, the official rule specifies **slot one**. We improvised a combined program at the event.

We also learned that our documentation had not been updated to the latest work and did not include the supporting images. The notebook records that the lack of up-to-date evidence affected the team's presentation and credibility.

25 September 2026

We made minor value adjustments during the limited time available between schoolwork and examinations.

29 September 2026

The team reorganised to make more complex but necessary decisions. We planned to revise the robot's structure to improve stability and to make sensor repositioning easier. More strategic sensor placement was intended to cover a larger detection area and improve movement around the mat.

We planned a rear-axle mechanism and additional LEGO gears so that the motor could operate with less effort and make better use of its capability. Work also continued on automatic recognition of the robot's orientation and whether it should move clockwise or counterclockwise. We planned further obstacle-detection experiments.

30 September 2026

The team reported progress on the previous day's work. We reinforced the structure and continued modifying the program. The notebook states that the remaining step was to observe the obstacle-detection code in action.

October / reinforcement and ERA

1 October 2026

We completed further structural modifications intended to reduce vibration at the connections. The motivation was that vibration could progressively affect the trajectory and cost points. We wanted straighter movement with fewer course disturbances.

2 October 2026

We finished the program modifications for the WRO Future Engineers competition in Ponce, changing the movement approach so that the robot could change direction more easily.

Naming the robot / ERA

We named the robot **ERA** to represent the beginning of a new generation of robotic technology and a new technological stage. Its eye-like ultrasonic sensor and bright colours inspired the idea of a “canvas of innovation”. Each detail represents the effort, creativity and dedication of the members, and what we can create together when we work towards a future goal.

Supporting evidence

- All software update groups and recent snapshots
- Six vehicle views and mechanical details
- Team and event photographs
- Original practice recordings
- Source notebook and corrected Word copy

The dated notebook entries provide the team's account. Project save timestamps, filenames and photograph dates are separate evidence sources; they are not used to infer undocumented test results or backdate repository history.